

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 01 — INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLERS

Student Edition

Authorized ICDFA training environments only
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Lab 01: Introduction to Programmable Logic Controllers

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 2–3 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the role of PLCs in industrial and critical-infrastructure environments.
- Identify core PLC components including processor, power supply, input modules, output modules, rack/chassis and expansion modules.
- Differentiate discrete (digital) and analog I/O.
- Launch OpenPLC Editor and inspect a ladder-logic motor-control example.
- Explain the purpose of normally open contacts, normally closed contacts and output coils.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Import the ICDFA-provided PLC VM into the virtualization platform specified by your instructor. Configure the laboratory network as Internal/Host-Only unless your instructor provides a different isolated configuration. Do not use Bridged networking.

Step 2. Create a clean snapshot named LAB01-START before making changes. Start the PLC VM and sign in with the assigned laboratory account.

Step 3. Confirm that the VM is using the expected training address 192.168.1.10. If your instructor supplied a revised image or address, record the value in your report instead of changing it without approval.

Step 4. Open the OpenPLC Editor from the desktop/application menu.

Step 5. Select File > Open and browse to the provided PLC Programming Language Examples directory. Open the Ladder Diagram project.

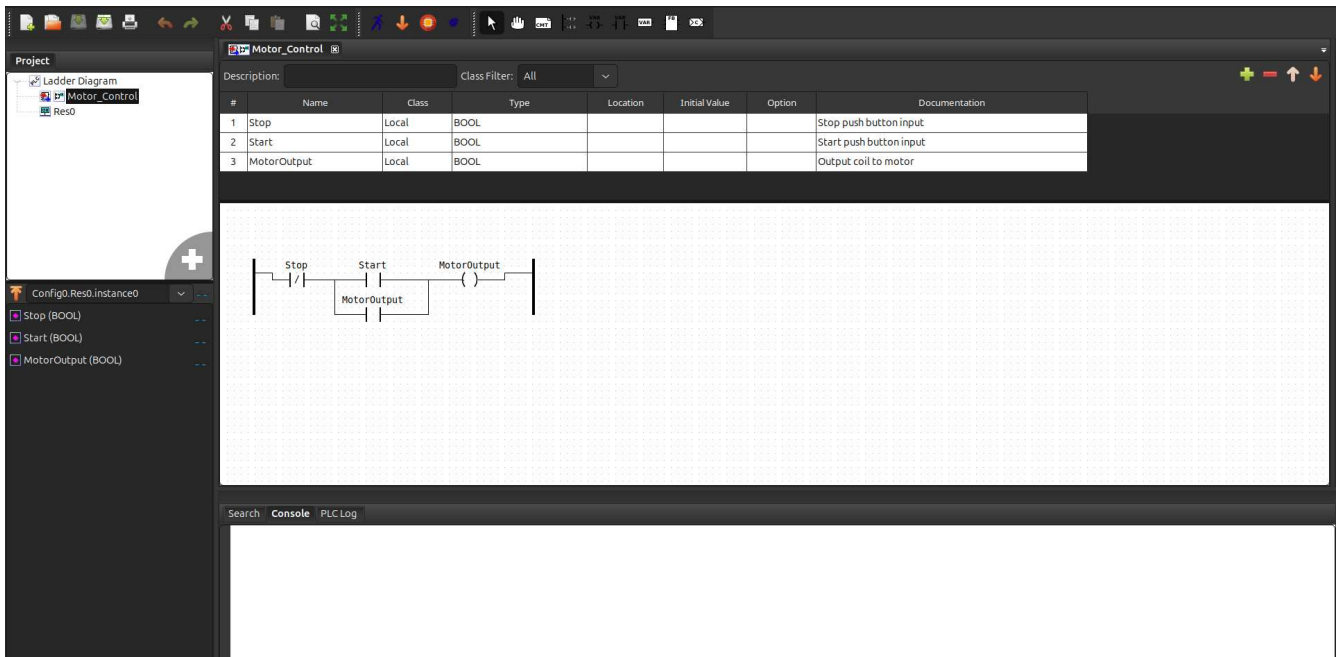
Step 6. Under the Project tree, open Res0 and edit instance0 (Motor_Control). Expand the ladder diagram so the full motor-control logic is visible.

Step 7. Identify the two power rails, the rung(s), the Start contact, Stop contact, MotorOutput contact used for latching, and the MotorOutput coil.

Step 8. Trace the logic from the left rail to the right rail. Explain what must be true for the motor output to energize and what causes it to de-energize.

Step 9. Record one example each of a discrete input, analog input, discrete output and analog output that could appear in an industrial process.

Step 10. Capture the required evidence, save your report, then restore or power off the VM as directed by the instructor.



Img: Screenshot showing the OpenPLC Motor_Control ladder diagram.

The logic shows a basic motor start/stop control. This control is for a system with a start button, a stop button and a motor. To turn the motor on, the start button is pressed which will energize the motor output.

In the logic, if the input for the NO (normally open) Start | | contact is true, then the MotorOutput () will be true. This will send an output signal to the motor in the field to turn it on. In this scenario, the Start input is temporary, therefore we must latch the MotorOutput () with a NO (normally open) set of contacts that reference the MotorOutput | |. This latch will be maintained until the Stop | / | input is received.

The Stop input is normally closed (NC), meaning that its true until an input is received. If the Stop input is received (when the stop button is pressed), then the input will be false which will break the circuit and de-energize the MotorOutput.

I/O Classification Table

Variable Name	Data Type	I/O Classification	Industrial Application Example
Emergency_Stop	BOOL	Discrete Input	A physical push-button switch that sends a binary on/off signal to the PLC to immediately halt a manufacturing conveyor belt.

Boiler_Pressure	REAL / INT	Analog Input	A pressure transmitter continuously measuring steam pressure inside a boiler, sending a variable 4-20mA signal representing a range of 0 to 100 PSI.
Pump_Contactor	BOOL	Discrete Output	A binary control signal sent from the PLC to a relay coil to switch a cooling water pump motor on or off.
Flow_Valve	REAL / INT	Analog Output	A variable 0-10V control signal sent by the PLC to a proportional control valve to dynamically adjust the aperture size and regulate fluid flow rate.

Reflection

Connecting PLCs to corporate networks completely transformed industrial operations, but it introduced a massive double-edged sword. On the operations side, the benefits are undeniable. Instead of a technician walking the plant floor with a clipboard to check machine statuses, engineers can now pull real-time telemetry into centralized SCADA systems. This makes predictive maintenance possible; an operator can monitor voltage drops or irregular motor behavior and catch a failing component before it halts an entire assembly line. It also allows for rapid remote troubleshooting, saving hours of costly downtime when a vendor or engineer can diagnose ladder logic faults without driving to the site.

However, bridging that traditional air-gap introduced severe cybersecurity realities. PLCs were fundamentally designed for physical safety and continuous uptime, not hostile network environments. They routinely lack basic security controls like encrypted communications, strong authentication, or the ability to process malformed network traffic without crashing. When you connect these devices to a wider network, you expose physical industrial processes to digital threats. A compromised IT jump host or engineering workstation can suddenly be weaponized to send unauthorized logic updates directly to the controller. We have moved from worrying primarily about mechanical wear and tear to worrying about an external adversary remotely overriding safety interlocks, forcing organizations to build heavy security controls like strict micro-segmentation around the PLC rather than relying on the controller itself.

Required Evidence & Submission

- Screenshot showing the OpenPLC Motor_Control ladder diagram.
- A labelled explanation of the Start, Stop, latch and MotorOutput elements.
- A completed I/O classification table with at least four industrial examples.
- A 150–250 word reflection on why network-connected PLCs create both operational benefits and cybersecurity considerations.

Recommended filename: <RegistrationNo>_LAB01_Introduction_to_Programmable_Logic_Controllers.pdf

Knowledge Check

1. What is the difference between an input module and an output module?

Response: The input module receives signals through input-sensing devices while the output module sends out a signal to load devices.

2. Why is a normally closed contact commonly associated with stop or emergency-stop logic?

Response: The normally closed contact is commonly associated with stop or emergency because it is in a true state until input is received. When the input is received, the state will become false thereby breaking the circuit and de-energizing the motor output.

3. Give one example of a discrete I/O signal and one example of an analog I/O signal.

Response: An example of a discrete signal is flipping the switch on/off while an example of an analog signal is 4-20 mA current loop measuring water tank level.

4. What operational risk can arise when industrial automation devices become network-connected?

Response: Operational risk that can arise when industrial automation devices become network-connected includes but is not limited to cyber-physical attack sabotages (malicious overrides, physical destruction) etc.

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.
Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.		